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Glasson et al.

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(54) **REMOTE SENSING DEVICE**

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E02D 29/14 (2006.01)
G01F 23/292 (2006.01)

(52) **U.S. Cl.**
CPC ***H01Q 1/225*** (2013.01); ***E02D 29/14*** (2013.01); ***G01F 23/2927*** (2013.01); *E02D 2600/10* (2013.01)

(58) **Field of Classification Search**
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(Continued)

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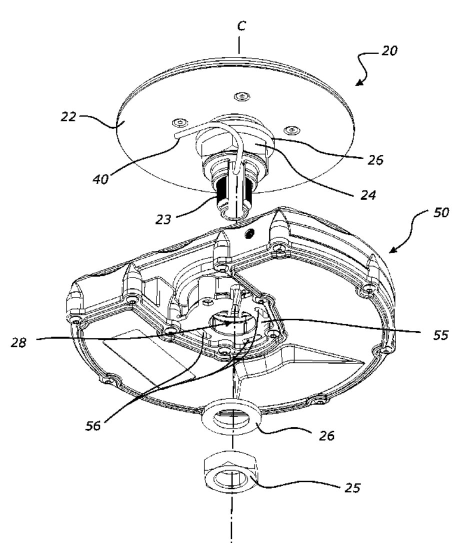
(57) **ABSTRACT**

A remote sensing system and device is provided for monitoring a condition within an infrastructure network. In particular, the system and device are configured to monitor components of civil water infrastructure such as sewers, rainwater infrastructure, freshwater infrastructure, manholes, catch pits, sewers, or outlets etc. The device includes an antenna unit having an elongate hollow shaft depending therefrom, and a monitoring unit for housing electrical components including a sensor for detecting a parameter. The monitoring unit is removably mounted to the antenna unit via the shaft, and connected to a component of the infrastructure being monitored, for example manhole cover or lid. The antenna unit and monitoring unit are separate and individually water-tight structures, and the elongate hollow connection shaft makes an essentially rigid connection, while finding a passageway between antenna unit and monitoring unit.

20 Claims, 11 Drawing Sheets



(58) **Field of Classification Search**
CPC ... E02D 2600/10; G01F 23/2927; H04Q 9/02; B65D 90/10; G01S 13/88; H05K 5/06
See application file for complete search history.

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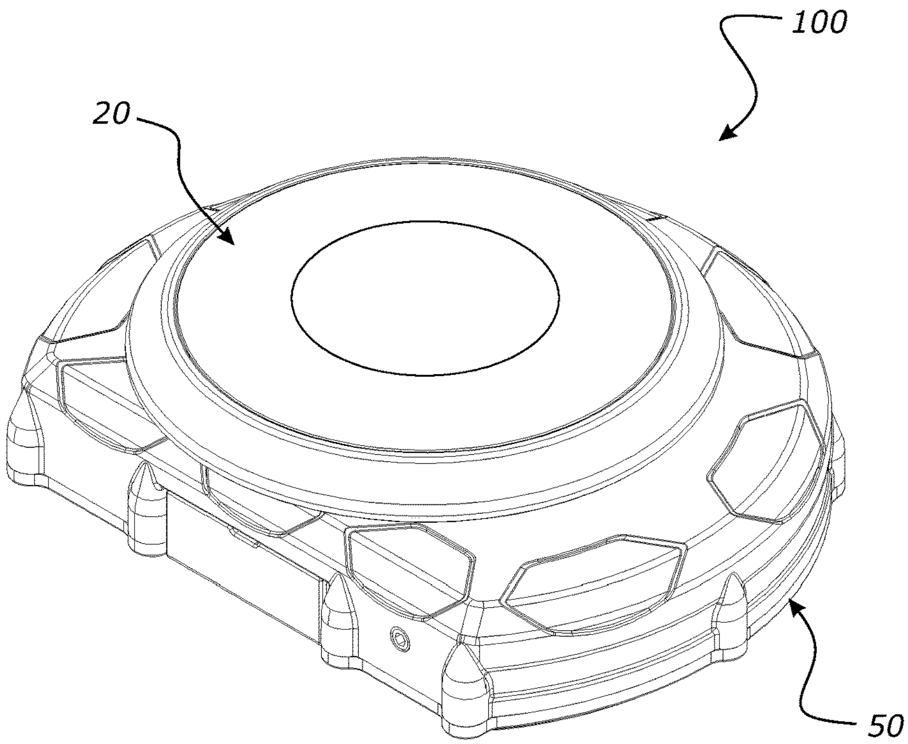

FIG. 1

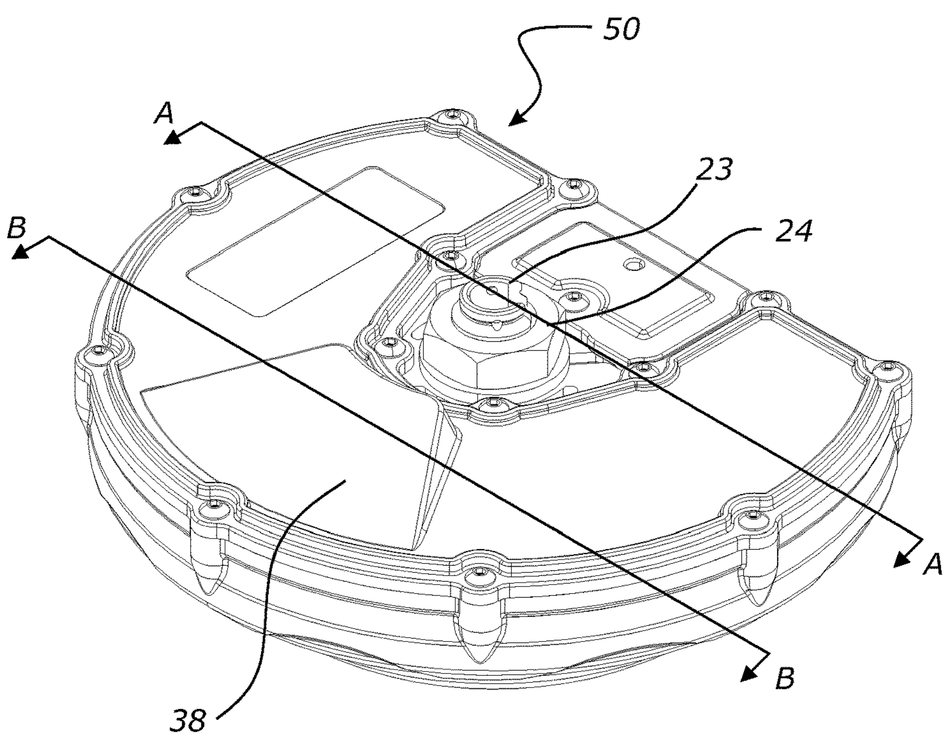

FIG. 2

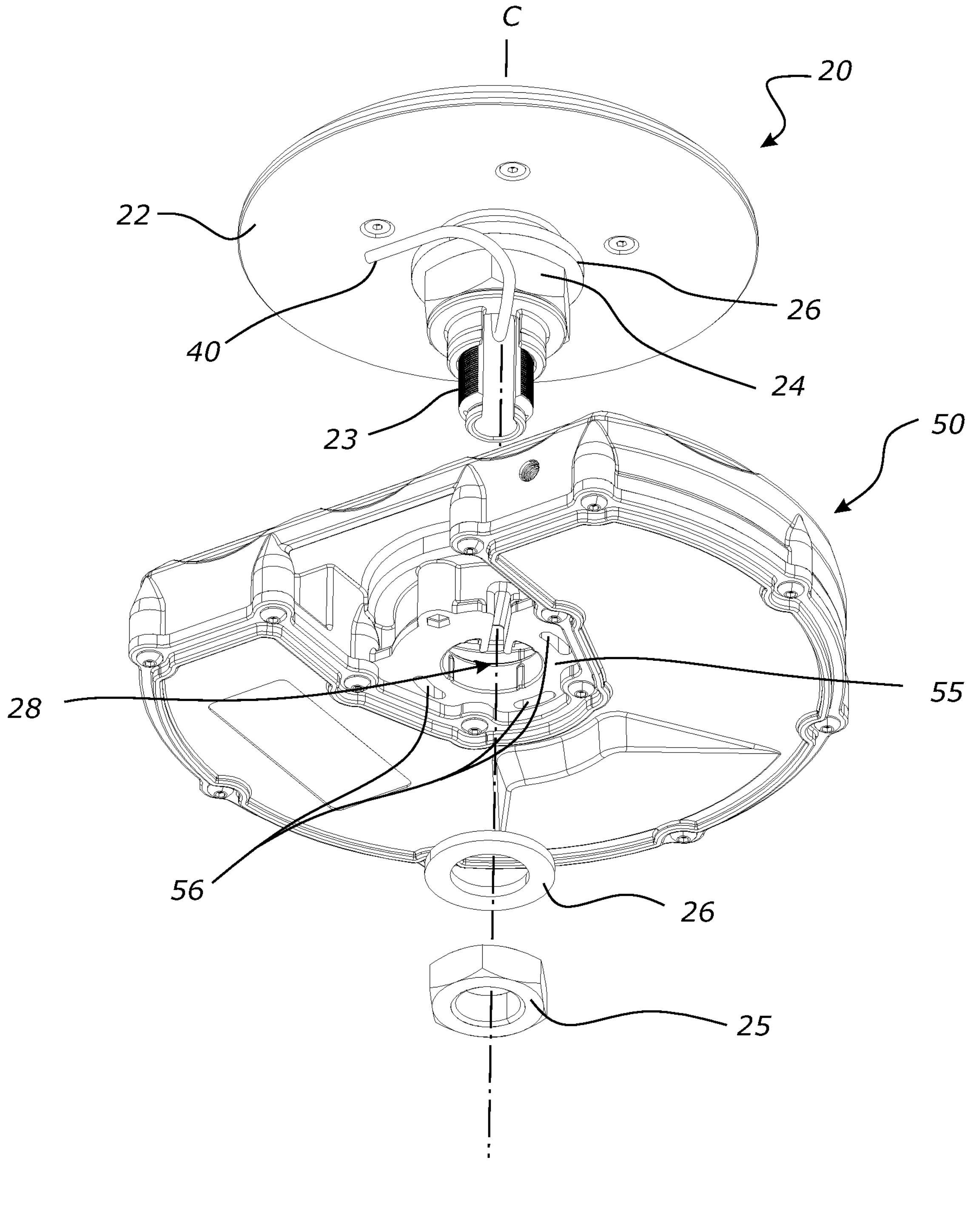

FIG. 3

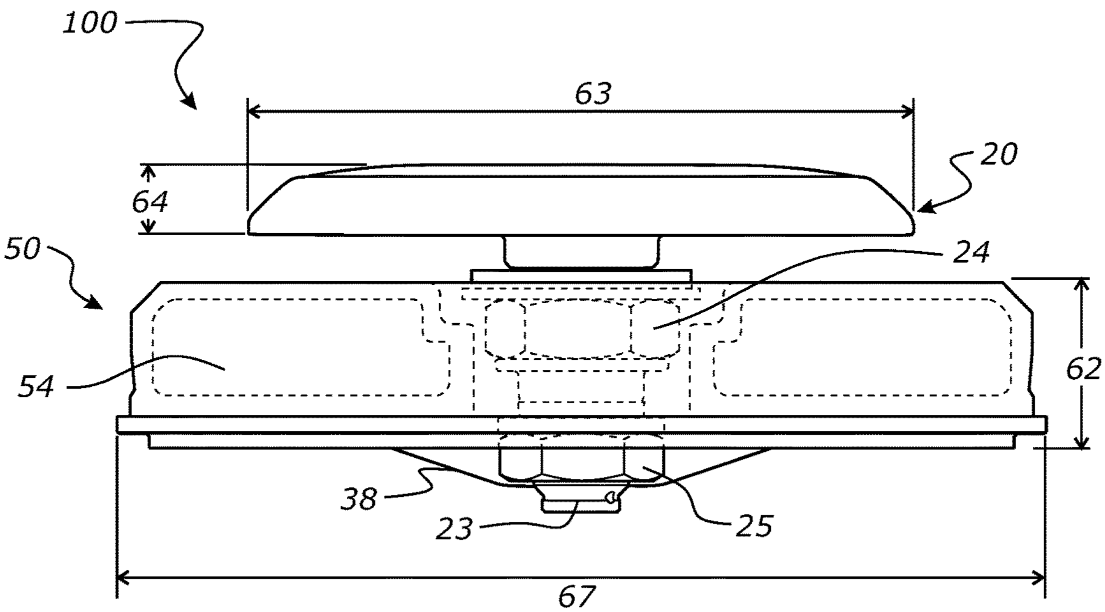

FIG. 4

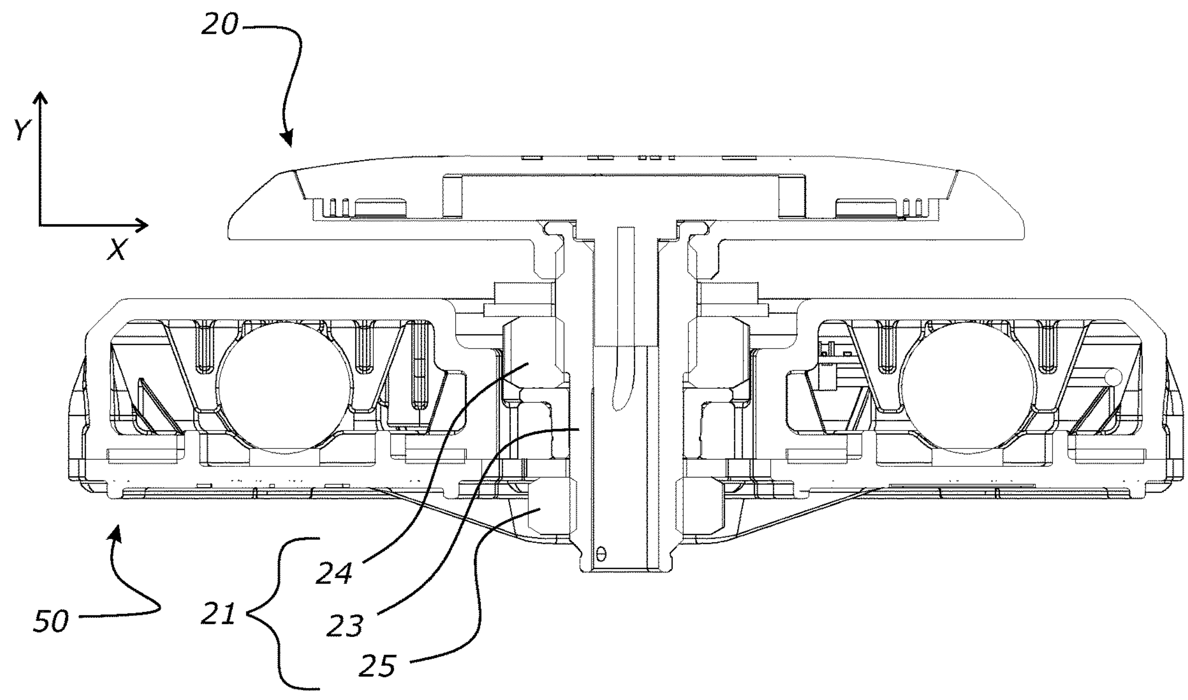

FIG. 5

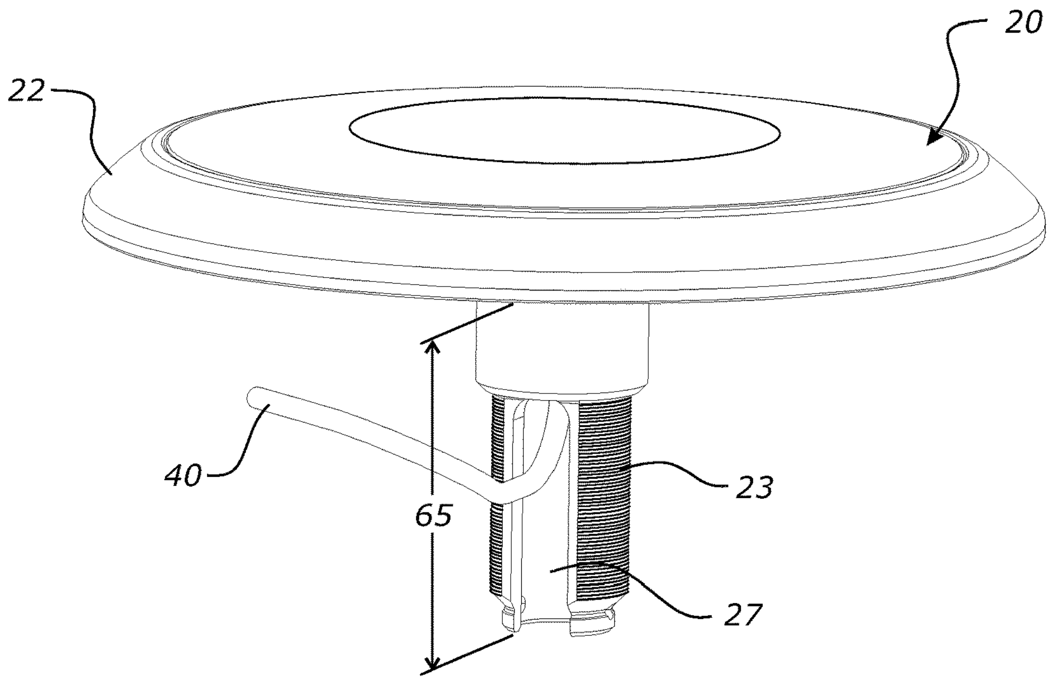

FIG. 6

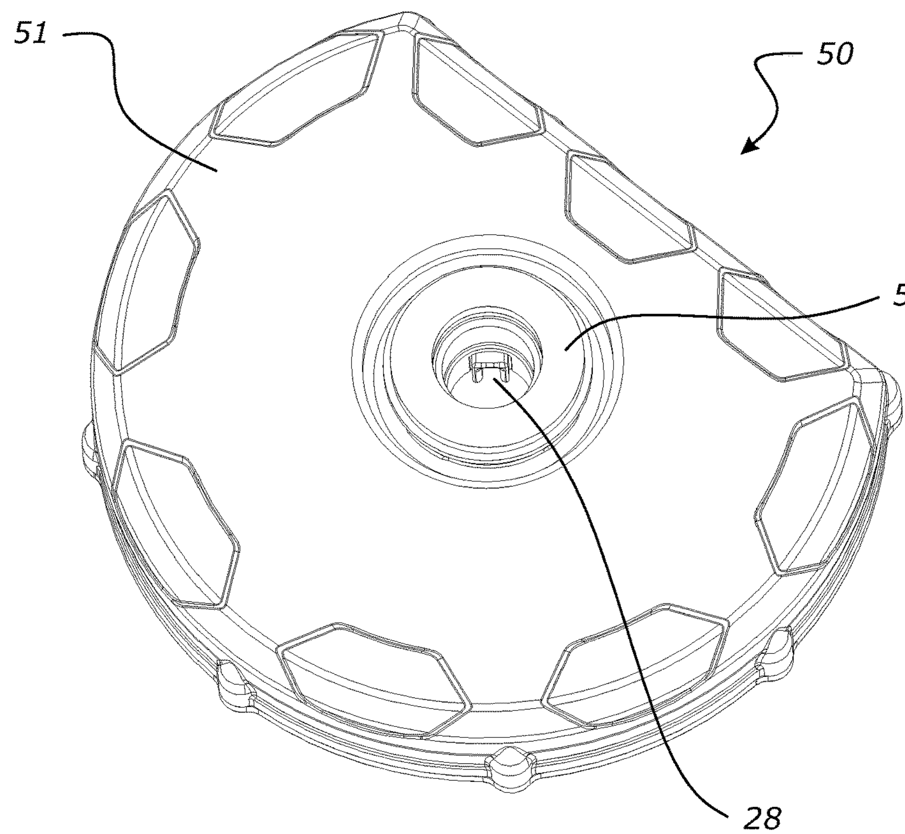

FIG. 7

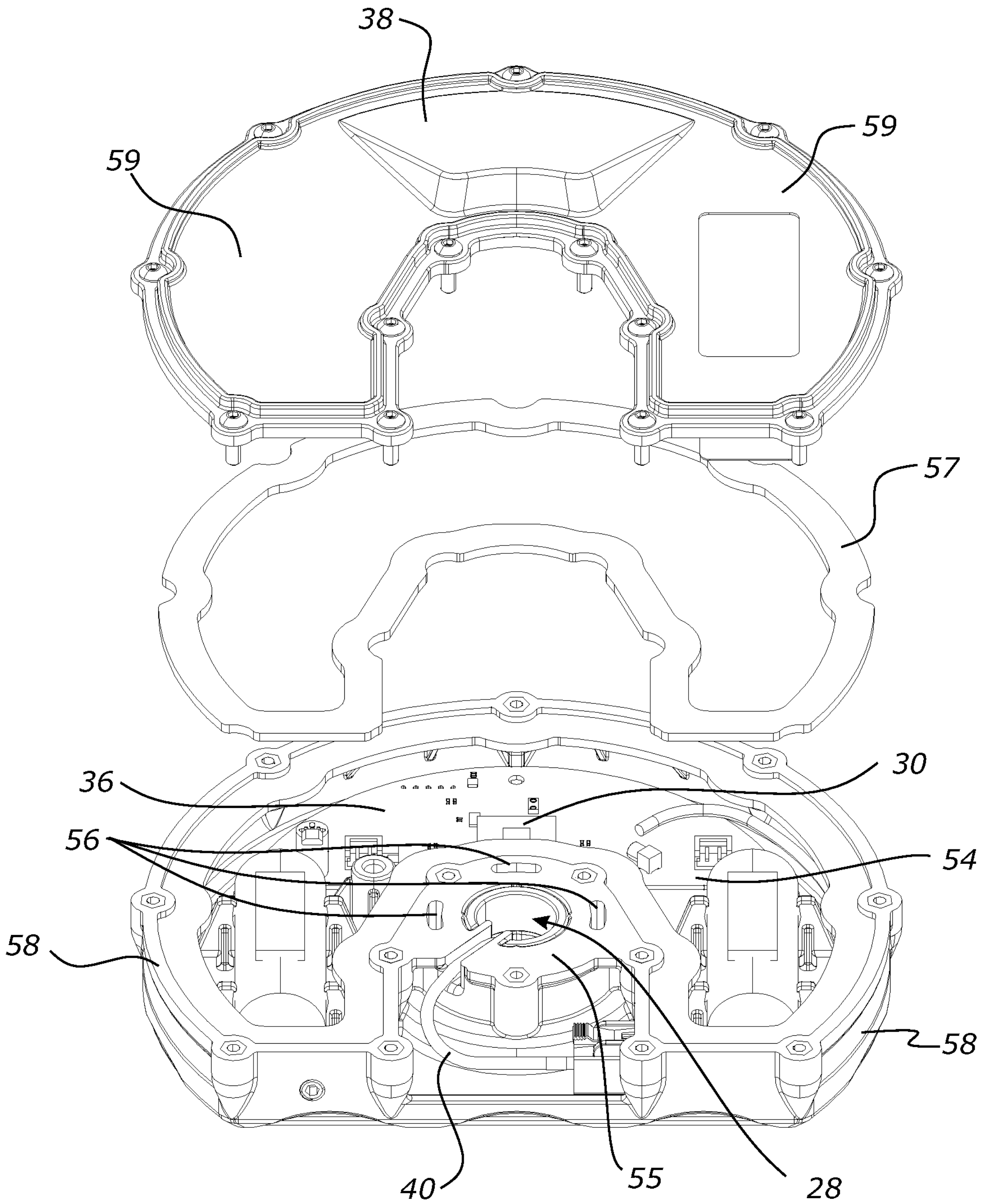

FIG. 8

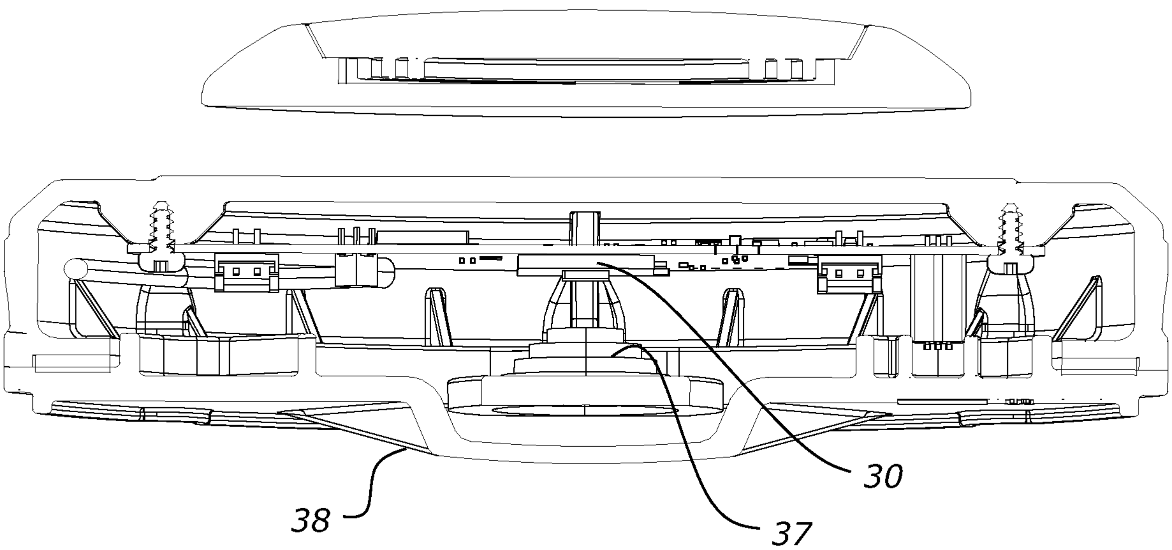

FIG. 9

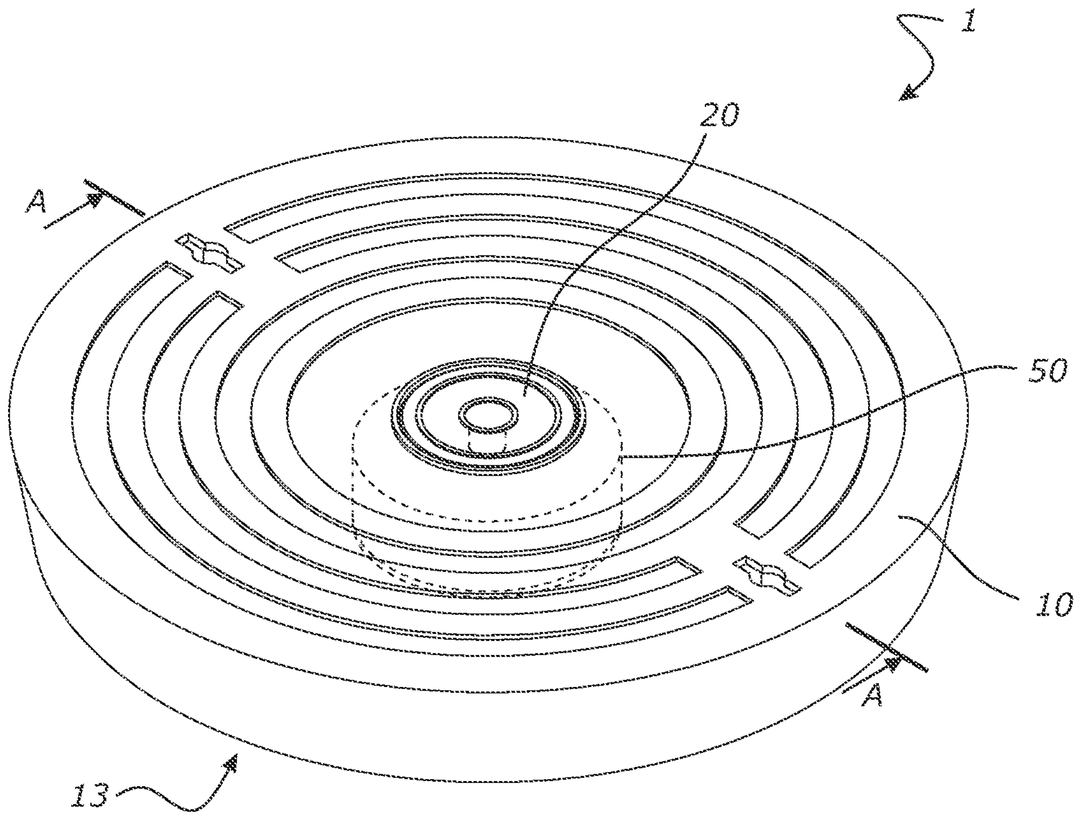

FIG. 10

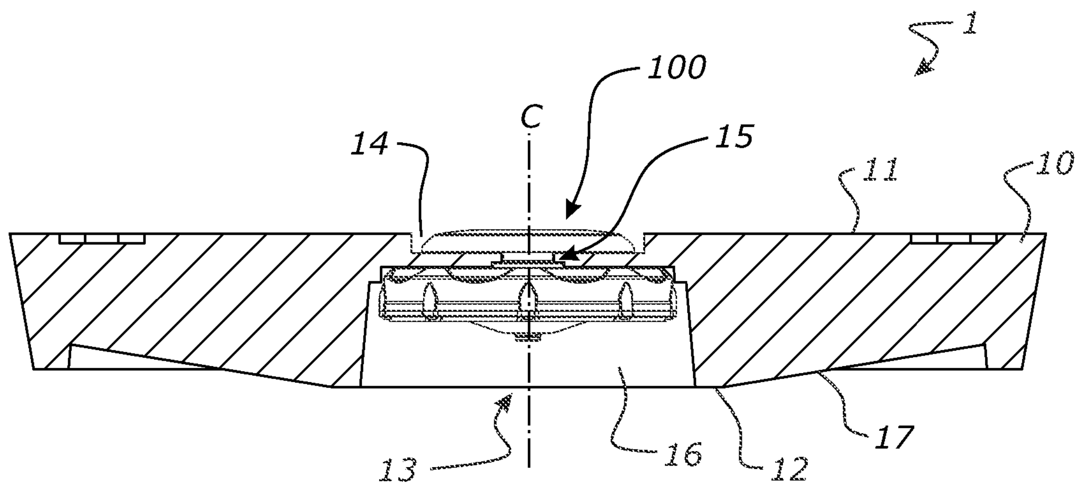

FIG. 11

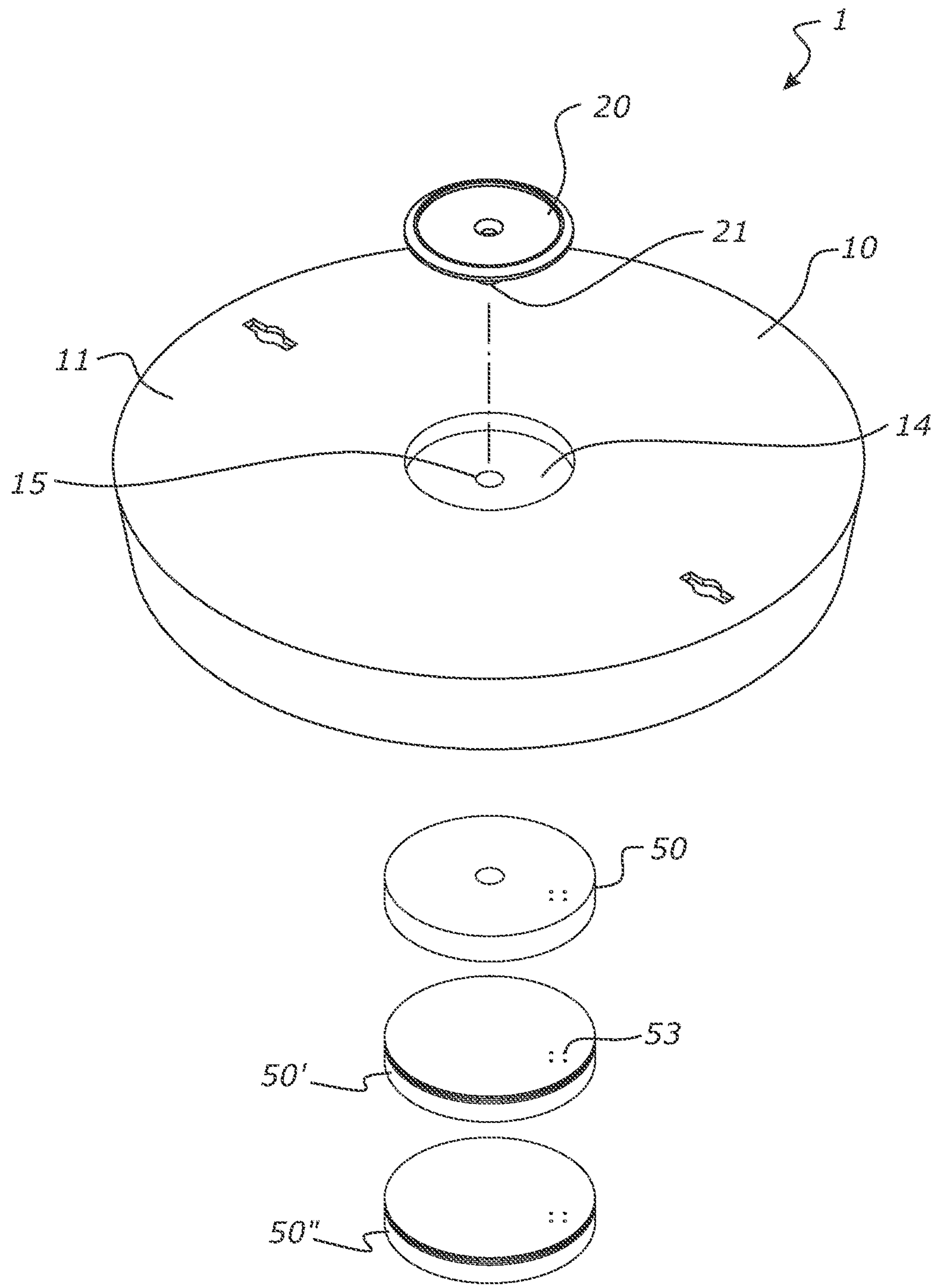

FIG. 12

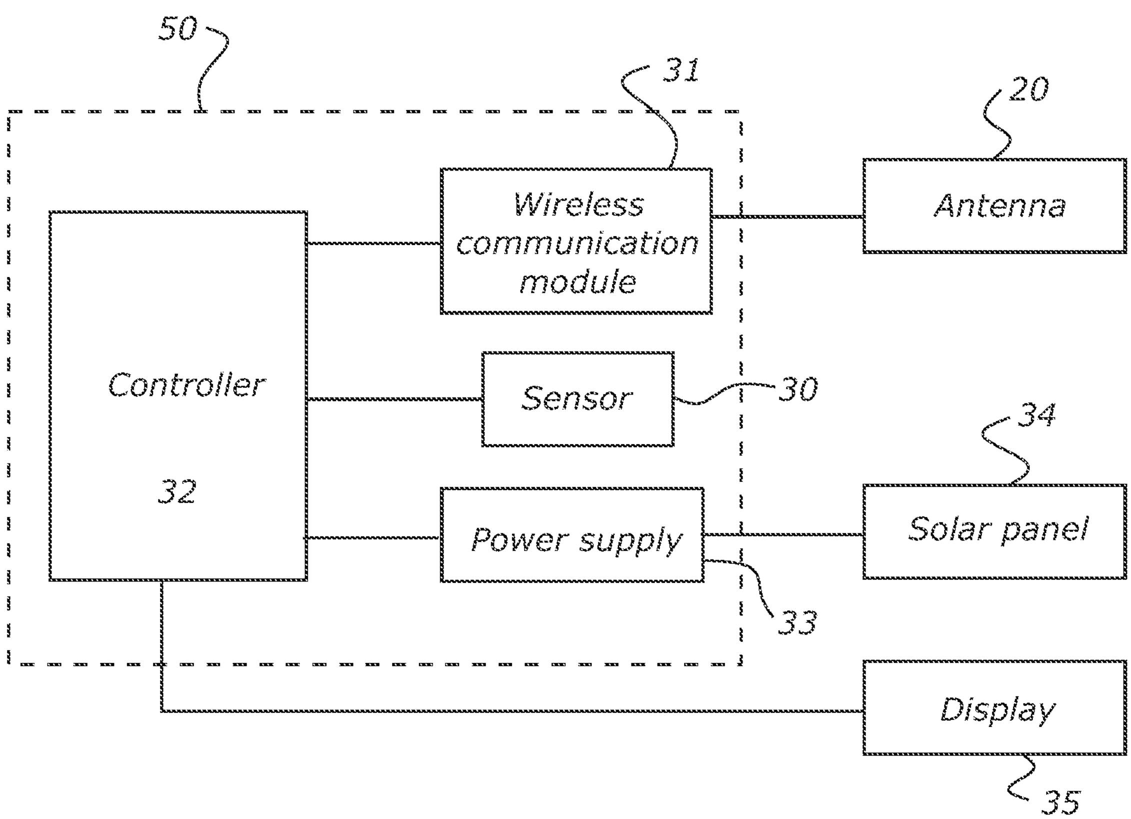
50
31
20
Wireless communication module
Antenna
Controller
Sensor
30
34
32
Power supply
Solar panel
33
Display
35

FIG. 13

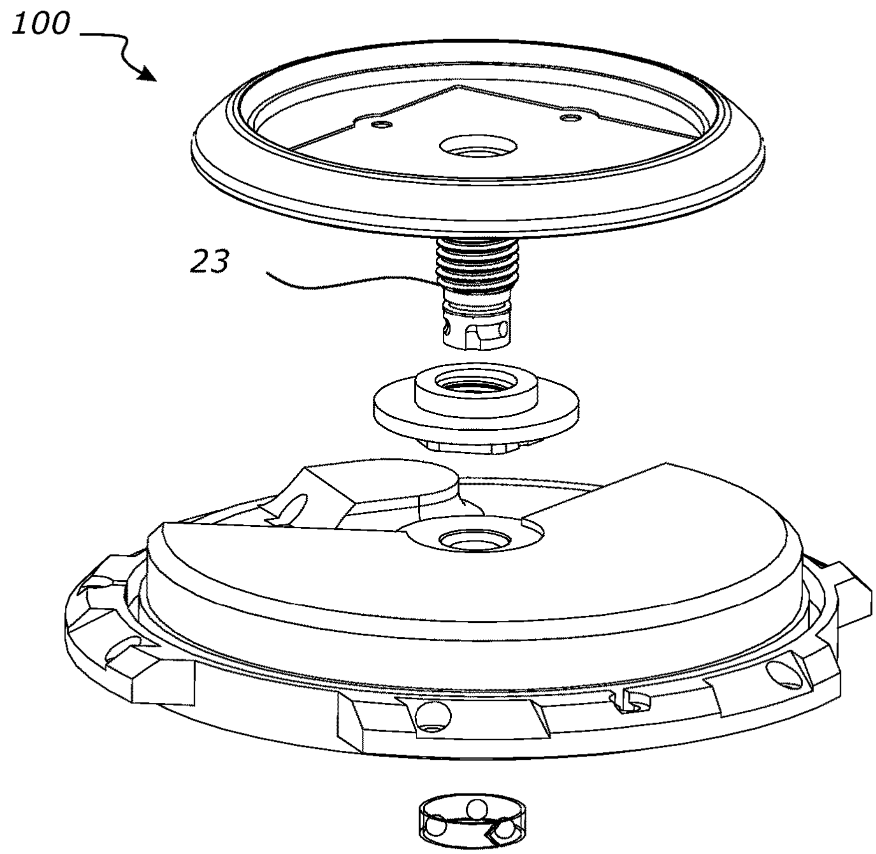

FIG. 14

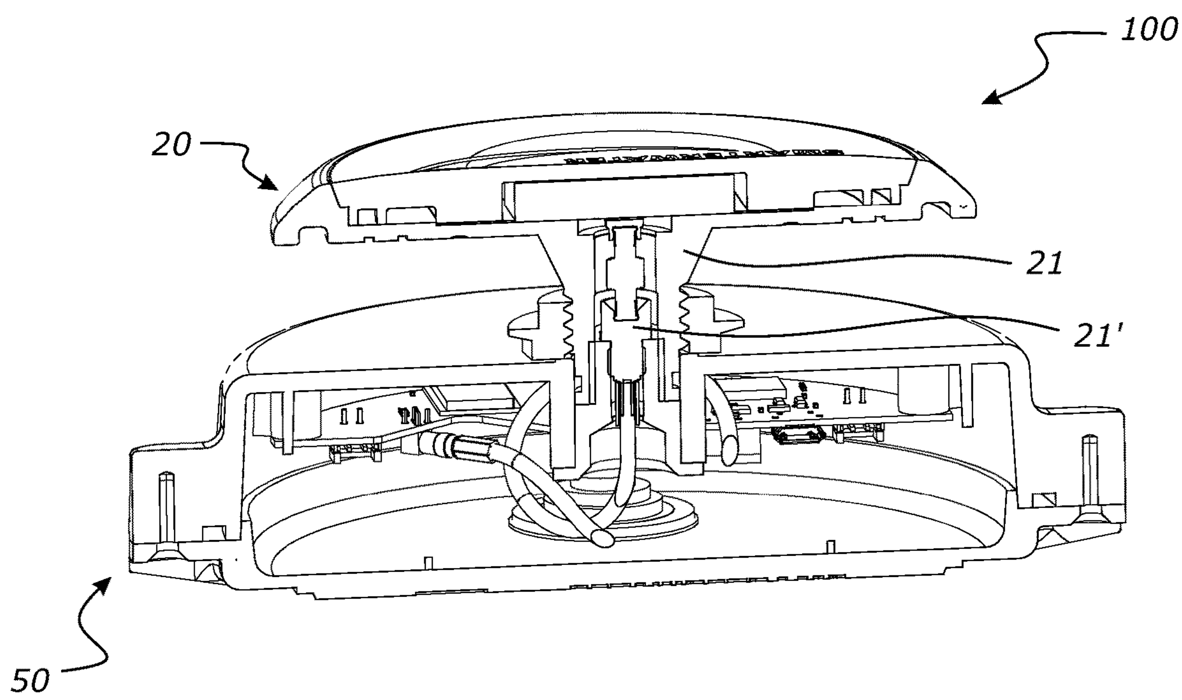

FIG. 15

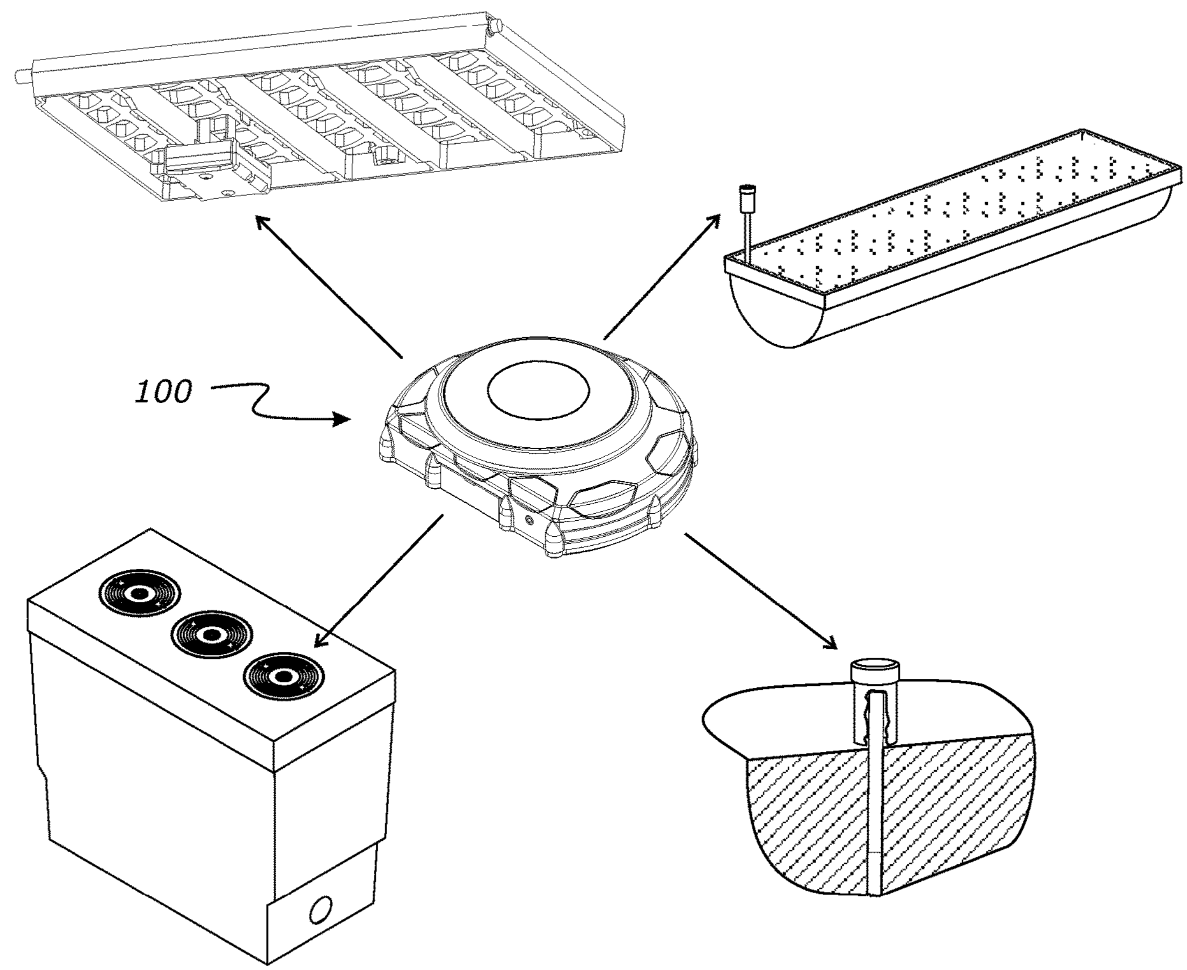

FIG. 16

REMOTE SENSING DEVICE

FIELD OF THE INVENTION

The present invention relates to a remote sensing device for monitoring a condition in water infrastructures such as manholes, sewers, stormwater catch pits, rain gardens, water troughs, gross pollutant traps, waste tanks, etc. More particularly, one preferred configuration relates to a remote sensing device for a smart manhole cover including one or more sensors integrated with the manhole cover.

BACKGROUND OF THE INVENTION

A manhole cover is a removable lid covering an opening of a manhole. Removing the manhole cover can provide access into the manhole. Access into manholes may be necessary for maintenance and servicing of underground infrastructure and facilities.

Monitoring parameters associated with the manhole such as water level, the presence of gas, gas concentration, temperature etc. may be useful. Traditionally, it has been necessary for operators to remove the manhole cover and enter the manhole to monitor or take measurements as required.

However, manhole covers are typically made from a heavy, solid material such as concrete, or cast metals. Removing the manhole covers and monitoring the manholes may therefore be difficult and may lead to injury to operators. Furthermore, entering into manholes may be dangerous due to exposure to toxic substances (e.g. H2S gas), or entering the manhole may create a fall hazard.

It may therefore be desirable to obtain information associated with a manhole, manhole cover or other water infrastructure as required remotely i.e. without going to the manhole or to other monitoring location, removing the manhole cover and accessing the manhole. It may also be desirable to provide a robust and durable smart manhole cover, suitable for use in its harsh and wet environment.

In this specification, where reference has been made to external sources of information, including patent specifications and other documents, this is generally for the purpose of providing a context for discussing the features of the present invention. Unless stated otherwise, reference to such sources of information is not to be construed, in any jurisdiction, as an admission that such sources of information are prior art or form part of the common general knowledge in the art.

For the purpose of this specification, where method steps are described in sequence, the sequence does not necessarily mean that the steps are to be chronologically ordered in that sequence, unless there is no other logical manner of interpreting the sequence.

It is an object of the present invention to provide a remote sensing device which overcomes or at least partially ameliorates some of the abovementioned disadvantages or which at least provides the public with a useful choice.

It is another object of the present invention to provide a smart manhole cover which overcomes or at least partially ameliorates some of the abovementioned disadvantages or which at least provides the public with a useful choice.

BRIEF DESCRIPTION OF THE INVENTION

According to a first aspect the invention broadly comprises a remote sensing device for monitoring a condition within an infrastructure network comprising:

an antenna unit having an elongate hollow shaft depending therefrom,

a monitoring unit for housing electrical components including a sensor for detecting a parameter, the monitoring unit removably mounted to the antenna unit via the shaft,

wherein the antenna unit and monitoring unit are separate and individually water-tight structures; and

wherein the elongate hollow connection shaft makes an essentially rigid connection to a component of the infrastructure being monitored.

According to another aspect the hollow connection shaft comprises an external thread for making the essentially rigid connection.

According to another aspect one of the securing elements is a nut associated with the threaded hollow connection shaft, the nut having large engaging surface for engaging against the component of infrastructure to form the essentially rigid connection.

According to another aspect the nut indirectly engages the component and the device includes one or more washers or gaskets.

According to another aspect the hollow connection shaft is connected directly to the component of infrastructure to form the essentially rigid connection.

According to another aspect an external thread of the hollow connection shaft engages directly with the component of infrastructure.

According to another aspect the hollow connection shaft is welded to the component of infrastructure.

According to another aspect the hollow connection shaft receives a cable connection.

According to another aspect an electrical connection is formed between the antenna unit and monitoring unit at the same time as a mechanical connection between the units is established with one motion.

According to another aspect the electrical connection is a push type electrical connector incorporated with the hollow elongate shaft.

According to another aspect the monitoring unit comprises a hub for mounting to the elongate shaft, and the hub includes an opening for receiving the hollow connection shaft to pass through and form the connection.

According to another aspect the monitoring unit is located in a region between the top and bottom ends of the hollow connection shaft.

According to another aspect the electrical components are located in a cavity of the monitoring unit, arranged to be spaced radially around the hub.

According to another aspect the monitoring unit comprises one or more drain ports located on the hub for water to pass through the monitoring unit, without entering the cavity housing the electrical components.

According to another aspect in plan view, the monitoring unit comprises one of the following general housing profiles:

a) donut,
b) ring,
c) partial-ring,
d) horseshoe.

According to another aspect the sensor is one of the following:

a) water level sensor,
b) non-contact flow sensor,
c) gas sensor,
d) temperature sensor,
e) moisture sensor,
f) tamper sensor,

g) vibration sensor, or

h) light sensor.

According to another aspect the sensor is a radar sensor comprising a lens.

According to another aspect the lens is integrated with a housing of the monitoring unit.

According to another aspect the monitoring unit comprises a water rejecting region in a focal area of the lens.

According to another aspect the water rejecting region is a ramp on a housing surface of the monitoring unit for draining water downwards away from the focal area.

According to another aspect the electrical components housed in the monitoring unit comprises one or a combination of:

a) a controller,

b) a wireless communication module,

c) a power supply.

According to another aspect all the electrical components are housed within one monitoring unit.

According to another aspect the antenna unit comprises a flat base for engaging with a flat surface of the component of infrastructure.

According to another aspect the antenna unit has a trafficable shape such as a dome shape.

According to another aspect the monitoring unit comprises an upper housing portion and a lower housing portion, the upper housing portion comprises downwards facing shell profile and the lower housing portion comprises a substantially flat lid profile.

According to another aspect a single gasket is provided between the upper housing portion and the lower housing portion to seal a mating surface between the housing portions.

According to another aspect the invention broadly comprises a smart manhole cover comprising:

a manhole cover body; and

a remote sensing device as described in the previous clauses;

wherein the remote sensing device is connected to the manhole cover body such that the antenna unit is located at an upper surface of the manhole cover body and the monitoring unit is located on an underside of the manhole cover body.

According to another aspect the elongate shaft passes through a through-hole of the manhole cover body for a connection between the antenna unit and monitoring unit.

According to another aspect the hollow connection shaft extends along a longitudinal connection axis passing through the through-hole.

According to another aspect the manhole cover body comprises a cavity on the underside of the cover for receiving the monitoring unit and the monitoring unit has a height less than the cavity height such that the monitoring unit does not protrude below a bottom surface of the manhole cover body.

According to another aspect the connection comprises one or a combination of the following connection types:

a) fasteners,

b) threaded,

c) snap-on,

d) twist-fit,

e) bayonet mount.

According to another aspect the connection comprises a double fastener stack including a primary fastener for first connecting the antenna unit to the manhole cover body and a secondary fastener for connecting the monitoring unit.

According to another aspect the invention broadly comprises a method of sensing a parameter comprising:

providing a remote sensing device as described in any of the previous clauses;

monitoring a parameter;

wirelessly communicating data from the remote sensing device to a remote receiver.

According to another aspect the method further comprises passing the hollow connection shaft through an opening of the monitoring unit and securing the monitoring unit with a hub fastener tightened against a surface of the monitoring unit.

According to another aspect the parameter is associated with a manhole and the remote sensing device is connected to a manhole cover body to form a smart manhole cover.

According to another aspect the hollow connection shaft connects the antenna unit and the monitoring unit on opposite sides of the manhole cover body.

According to another aspect the method further comprises securing the antenna unit to the manhole cover body before securing the monitoring unit.

According to another aspect the method further comprises positioning the monitoring unit onto the hollow connection shaft on the underside of the manhole cover body and securing the monitoring unit toward and/or against a bottom surface of the cover.

According to another aspect the monitoring unit is clamped between the manhole cover body and a nut.

According to another aspect the method further comprises rotating one or both of the antenna unit and the monitoring unit to their operating position before securing the units in position.

According to another aspect the antenna unit is secured to the manhole cover body before the monitoring unit is secured to the manhole cover body.

According to another aspect the remote sensing device is retrofitted to an existing manhole cover body.

According to another aspect the remote sensing device is connected to the manhole cover body during manufacture.

According to another aspect the remote sensing device is associated with and connected to one of the following:

a) sewers,

b) stormwater catch pits,

c) rain gardens,

d) water troughs,

e) gross pollutant traps,

f) waste tanks.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention will now be described by way of example only and with reference to the drawings in which:

FIG. **1** shows a top perspective view of a remote sensing device.

FIG. **2** shows a bottom perspective view of the remote sensing device.

FIG. **3** shows an exploded view of the remote sensing device.

FIG. **4** shows a side view of the remote sensing device.

FIG. **5** shows a cross section of the remote sensing device along line AA of FIG. **2**.

FIG. **6** shows a perspective view of an antenna unit and hollow connection shaft.

FIG. **7** shows a perspective view of a monitoring unit.

FIG. **8** shows an exploded view of the monitoring unit.

FIG. **9** shows a cross section of the remote sensing device along line BB of FIG. **2**.

FIG. **10** shows a top perspective view of a smart manhole cover.

FIG. **11** shows a side view of the smart manhole cover.

FIG. **12** shows an exploded view of a smart manhole cover with a plurality of modular units.

FIG. **13** shows a simplified diagram of components of the remote sensing device.

FIG. **14** shows another smart manhole cover embodiment having a bayonet connection.

FIG. **15** shows another smart manhole cover embodiment having a quick connect coaxial connection.

FIG. **16** shows the remote sensing device installed in different applications.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

The present invention is a remote sensing device **100** for monitoring a condition within a water infrastructure network as illustrated in FIGS. **1** to **16**. In the preferred configurations, the remote sensing device **100** is for a manhole cover for covering an opening of a manhole. The remote sensing device **100** is configured to monitor manholes or other water infrastructure where suitable with no or minor modification to the described invention (e.g. sewers, stormwater catch pits, rain gardens, water troughs, gross pollutant traps, waste tanks etc.) and wirelessly communicate data to a remote receiver.

Generally, water infrastructure like manholes are harsh and wet environments. The remote sensing device **100** has delicate components to achieve certain performance requirements (e.g. good signal transmission, sensor able to obtain good readings). Preferably, the present invention can provide a simple, repeatable and robust way of mounting these delicate components to a manhole cover while also protecting the components against the harsh and wet environment it is installed in.

A manhole cover is often placed on and/or dragged across the ground during installation. Consequently, any components protruding out the underside of the manhole may be easily damaged. It may therefore be desirable to provide a low-profile device and protect components on the underside of the device. Specific components in the device, structures and arrangements are described below to provide the desired performance capabilities, while also providing a durable and robust device.

Furthermore, water infrastructure/manhole covers can have different profiles (e.g. different cavity and/or rib configurations on the underside of the manhole cover) or are formed from different material (cast iron, metal, composites, plastic, concrete etc.). Repeatable and flexible installation on different structures may be desirable so that the device can be installed onto a wide of range of different cover profiles. Installing the remote sensing device **100** to traditional infrastructure can quickly convert the structure to 'smart' infrastructure with monitoring and communication capabilities. The remote sensing device **100** of the present invention provides structures which can be simply installed onto different water infrastructure/manhole covers with different profiles with no or limited modification to the cover. The remote sensing device **100** may also be simply removed for replacing, upgrading and/or maintaining components as required.

The general structure of the various configurations of the remote sensing device **100** and method of sensing a parameter using the device as shown in the figures will now be described.

It will be appreciated that these figures illustrate the general principles of construction, and that the invention is not limited to the precise mechanical configurations illustrated.

Device Components

As shown in FIG. **1**, there is a remote sensing device **100**. The remote sensing device **100** comprises an antenna unit **20** and a monitoring unit **50**, removably mounted to the antenna unit, for housing electrical components.

Preferably, the antenna unit **20** and the monitoring unit **50** are separable and individually water-tight structures. The individually water-tight units allow the structures to be separated and easily installed onto the infrastructure it is monitoring from.

The antenna unit **20** is a structure which carries an antenna (i.e. an antenna is embedded in the antenna unit **20** structure). The antenna is configured to send or receive a signal to a remote receiver outside the manhole/from the monitored infrastructure.

Preferably, the antenna unit **20** comprises a flat profile with a relatively large surface area for improved signal.

Preferably, the antenna unit **20** is located on an upper region and/or outside a manhole/infrastructure being monitored (i.e. on top of a manhole cover), for improved signal transmission.

In the preferred configurations, an antenna wire is moulded or otherwise embedded into the antenna unit **20** (e.g. an antenna patch embedded in the antenna unit).

The antenna unit **20** is exposed on the top surface of the manhole cover and therefore exposed to impact from traffic and may be subject to harsh and wet conditions. In some configurations, the antenna patch is embedded in a solid or substantially solid housing (potting to make a solid or near solid unit). A solid and robust antenna unit **20** structure can reduce potential leak pathways which can damage the internal antenna and/or reduce performance issues or damage due to impact/vibrations due to vehicles passing over the manhole cover.

In the preferred configurations, the antenna unit **20** is an independently watertight structure from the monitoring unit **50**, for housing and protecting the antenna (i.e. the antenna unit **20** is watertight even when it is not connected to the monitoring unit **50**). The units are preferably watertight, to protect against the harsh environment it is in (e.g. where liquids and/or dangerous gases are present). A watertight housing may be understood to be a protective housing to limit or prevent gas/liquid immersion into the internal cavity of the unit which houses the antenna/electrical components.

In the preferred configurations, the materials and/or components incorporated in the remote sensing device **100** is rated for potentially explosive environments. For example, the materials and/or components are adapted to prevent minimise likelihood of fire or explosion hazards when potentially in contact with combustible or flammable gases, vapours etc. The housing and structure of the device is preferably suitably airtight to reduce the potential entrance of dangerous gases.

The monitoring unit **50** houses electrical components such as one or more sensors **30**, a controller **32**, a wireless communication module **31** (i.e. driving the transmitting and receiving protocol via the antenna), and/or power supply **33** (e.g. battery).

In the preferred configurations, the monitoring unit **50** is an independently watertight structure from the antenna **20**, for housing and protecting the electrical components (i.e. the monitoring unit **50** is watertight even when it is not connected to the antenna unit **20**).

It should be appreciated that by having the electrical components housed within the monitoring unit **50**, wires or protuberances outside the housing are eliminated or at least reduced. Consequently, the monitoring unit **50** can provide protection to the components it houses, in the harsh environment of a manhole and/or during transportation, installation and removal of the manhole cover.

In the preferred configuration, the remote sensing device **100** is installed/connected to a manhole cover body **10** to form a smart manhole cover **1**, as shown in FIGS. **10** and **11**. Simply connected the remote sensing device **100** can transform a traditional manhole cover to a smart manhole cover **1** with monitoring and wireless communication capabilities.

The smart manhole cover **1** comprises a manhole cover body **10** for covering an opening of a manhole. In some configurations, the cover body **10** is an existing manhole cover and the remote sensing device **100** for monitoring and/or communication information from the manhole can be simply and easily retrofitted to the existing manhole cover body. In other configurations, the components are connected to a cover body **10** during manufacturing of the smart manhole cover **1**.

Preferably, once the remote sensing device **100** is rigidly connected to the manhole cover and the device is incorporated with/into the manhole cover, it becomes part of the manhole cover.

Preferably the remote sensing device **100** is integrated with a manhole cover body **10**. Where the device is integrated with the cover body **10**, it should be understood as being connected to the cover such that it becomes part of the cover and will move with the cover (e.g. as the cover is removed or placed over the manhole). Preferably, integrated with the cover body can be understood to mean that the device is located on the manhole cover and/or secured to the manhole cover so that it remains on the manhole cover upon impact on the cover and when it is moved (e.g. dragged into position). The remote sensing device **100** can be integrated/connected with the cover body **10** by direct or indirect coupling.

It should be appreciated, in these preferred configurations where the device is integrated with the manhole cover, the components for monitoring and communicating information to/from the manhole is on/within the lid. Therefore, an operator does not need to enter the manhole to take measurements or enter the manhole to install the sensor down into the manhole (improving safety of installation). A smart manhole cover **1** may be simply manufactured and put in place at the manhole site. Alternatively, the smart manhole cover **1** may be simply formed on the side of a road by connecting the remote sensing device **100** to an existing manhole cover with no or minor modifications (e.g. forming a through-hole **15**) to the cover to form the smart manhole cover.

Further, installation/connection of the components to the manhole cover or other infrastructure is simplified as the units as a whole is connected to the cover rather than individual components. Simplifying the installation process of components to the manhole cover reduces time and costs of forming the smart manhole cover, whether it be by retrofitting the components to an existing manhole cover or manufacturing a smart manhole cover. Simplifying the installation process also reduces the skill level/expertise required to install the remote sensing device on-site (e.g. when retrofitting to an existing cover) or during manufacture.

In other configurations, the remote sensing device **100** is secured using any suitable method to different infrastructure such as a catch pit as shown in FIG. **16**. The device **100** is secured to the catch pit grate, to monitor parameters e.g. water level below the catch pit.

In another configuration, the remote sensing device **100** is incorporated in a smart rain garden system (as shown in FIG. **16**), to measure parameters such as moisture in different media levels in a rain garden. The device **100** can be secured to a structure, such as a pipe inserted into the rain garden media.

In another configuration, the remote sensing device **100** is incorporated into a smart trough system (as shown in FIG. **16**), to measure water level inside a water trough. The device **100** can be secured to a structure, such as a pipe or post located above the water level of the trough.

In yet configuration, the remote sensing device **100** is incorporated into a tank (e.g. a waste tank) (as shown in FIG. **16**), to measure water level inside. The device **100** can be secured to a structure, such as on the cover or upper surface of the tank.

In some configurations, all the electrical components (e.g. sensor **30**, the controller **32**, the wireless communication module **31**, and the power supply **33**) are housed within one monitoring unit **50**. In other configurations one or more monitoring units **50** house the electrical components.

The antenna unit **20** and monitoring unit **50** each provides a water-tight housing for the antenna or the other electrical components. The units **20**, **50** are watertight such that the housing/structure protects the internal components from water damage. Preferably, each structure/unit **20**, **50** is hermetically sealed and/or includes sealing features (e.g. gaskets) to eliminate or at least reduce potential leak pathways into the internal components.

It should be appreciated separate structures for the antenna unit **20** and the monitoring unit **50** can provide advantages as they can be simply installed on different manhole covers (e.g. with different cavity and rib configurations on the underside of the cover).

Further, the antenna and monitoring units **20**, **50** can installed on opposite sides of a manhole cover **10** and removed for individual maintenance or replacement as required. For example, the antenna unit **20** may be replaced to accommodate different wireless transmission technologies and/or frequency depending on the type of radio communication technology used. Monitoring units **50** with different functionality may be provided as required to the water infrastructure.

Further, the antenna unit **20** can be located at an upper surface **11** of the manhole cover to send or receive a good signal outside the manhole (improved signal propagation as the antenna is located on top of the manhole cover). The monitoring unit **50** can be located on an underside **13** of the manhole cover so that sensor **30** is within or at least located towards the inside of the manhole to detect a parameter or condition within the manhole and the electrical components are protected from the elements above ground.

Preferably, the antenna unit **20** is located at or towards the top surface **11** of the cover body **10** (for signal transmission).

In one configuration, the antenna unit **20** is located in a recess **14** on the top surface of the cover body as shown in FIG. **11**. In these configurations a top surface of the antenna unit **20** is on the same plane or below the top surface **11** of the cover body **10**. Preferably, the recess **14** is suitable for trafficability over the manhole cover. For example, the recess **14** preferably has a depth equal or less to 13 mm.

In another configuration, the antenna unit **20** is located on a top surface **11** of the cover body **10** without a recess **14**.

In these configurations, the low profile of the antenna unit **20** is sufficient to not protrude too far above the manhole cover **10** as illustrated in FIG. **10**.

The antenna unit **20** comprises a flat base **22** for receiving the antenna. In some configurations, the antenna unit **20** has a trafficable shape (for pedestrians and vehicles to pass overhead without damaging the antenna unit) e.g. a dome shape as best shown in FIG. **6**. Preferably, a top perimeter of the antenna tapers downwards (e.g. towards the bottom of the manhole cover as best shown in FIG. **11**).

It should be appreciated, the recessed or low-profile of the antenna unit **20** can provide advantages such as reducing the risk of the device becoming a hazard for pedestrians or vehicles passing over the smart manhole cover and also minimises damage to the components of the smart manhole cover due to potential shear forces or vibrations passing through to the device.

Compact Cover

In the preferred configurations, the components of the smart manhole cover **1** are arranged to form a compact/low-profile cover. Preferably any protrusion above and below the manhole cover is minimised or eliminated (i.e. minimal protrusion above or below the top surface **11** and bottom surface **12** of the cover body **10**).

A typical manhole cover has limited space on the underside of the cover (due to a limited cavity height **16** and obstructions e.g. ribs **17**). Specific structures and arrangements of the structures of the present invention uses the limited space efficiently so that the device is compact/low-profiled, rigidly secured to the cover while providing the desired monitoring/communication performance requirements.

In configurations where protrusion of the remote sensing device **100** above the plane of the top surface **11** of the cover body **10** is minimised, the manhole cover can provide the advantage of reducing the likelihood of being a hazard to vehicles and/or pedestrians above the cover and damage to the antenna.

The manhole cover body **10** comprises a cavity **16** on the underside **13** of the cover to receive the monitoring unit **50** as referenced in FIG. **11**.

In configurations where protrusion of the remote sensing device **100** below the plane of the bottom surface **12** of the cover body **10** is minimised or eliminated, operators may move, drag or place the manhole cover **1** on the ground without damaging the components/unit **50** on the underside **13** of the cover body **10**. The bottom surface **12** (e.g. ribs **17**) of the cover body **10** would contact the ground rather than components of the manhole cover **1**. This can improve the longevity of the components connected to the manhole cover, minimising costs associated with maintenance and replacement of components.

Preferably, the monitoring unit **50** has a height less than the height of the manhole cover body **10**/cavity **16** so that the remote sensing device **100** does not protrude below the bottom surface **12** of the cover body **12**.

In other configurations, the lowest region of the smart manhole cover **1** is a support element extending from the remote sensing device **10** (e.g. a shaft/nut extending from the antenna unit **20**) to contact the ground. In these configurations, the manhole cover rests on an edge of the cover and the extending support element so that the monitoring unit housing **51** does not contact the ground and damage to the components may be minimised.

To achieve a low-profile cover, particular components are provided in the remote sensing device **100** and are arranged and connected to the manhole cover **10** in specific configurations as described in more detail below.

Main Connection

In the preferred configurations, the remote sensing device **100** comprises a connection **21** configured to provide a rigid, stable and strong connection to secure/mount the device to the manhole cover body **10** or infrastructure (so that that the device becomes part of the manhole cover or other structure). It should be appreciated a strong connection can improve the lifespan and/or performance of the components of the remote sensing device **100**. By providing a strong and rigid connection between the antenna and monitoring units **20**, **50** and the manhole cover **10**, loosening/movement of the components from its operating position, (or other damage) from vibrations/shear forces due to vehicles and pedestrians passing over the manhole cover may be minimised. A robust and reliable connection is desirable given the potential high forces (especially due to heavy vehicles, including airplanes) and frequency of traffic (high instantaneous and/or cyclic loads due to frequent traffic can loosen or break components/the housing) of the remote sensing device **100**.

In the preferred configurations, the connection of the device **100** is simple, requiring no or limited modification to traditional manhole covers.

Preferably, the remote sensing device **100** is simply connected to a manhole cover body by inserting a hollow elongate shaft **23** depending from the antenna unit **20** through a through-hole **15** in the manhole cover and securing the antenna unit to the cover body (e.g. with a large fastener or other securing techniques discussed below).

Preferably, the hollow elongate shaft **23** makes an essentially rigid connection to a component of the infrastructure being monitored (e.g. connected to manhole cover lid). The connection is essentially rigid such that the device is sufficiently connected to a component/structure of the monitored infrastructure network and operate for a desired time. Once installed through the manhole cover the elongate shaft **23** provides a mounting structure on the underside **13** of the manhole cover for receiving and supporting the monitoring unit **50** and/or other structures and units on the underside of the manhole, to provide the desired functionality for the remote sensing device **100**.

It should be appreciated that once the connection has been established (i.e. inserting the elongate shaft **23** through the manhole cover and securing the antenna unit **20**), the traditional manhole cover is ready to receive the monitoring unit **50** and has been converted to a smart manhole cover **1** with monitoring/wireless communication capabilities.

In some configurations, the manhole cover comprises an existing through-hole **15** passing through a top and bottom surface **11**, **12** of the cover body **10** (referenced in FIGS. **11** and **12**). In other configurations, through-hole **15** can be easily created (either on-site or off-site) to allow the connection **21** to passes through the manhole cover.

It is anticipated the remote sensing device **100** with a separable antenna unit **20** and monitoring unit **50** has the flexibility of being installed on different manhole covers with different cavity **16**/rib **17** configuration on the underside of the cover. Preferably, the through-hole **15** is formed in a region with sufficient cavity **16** space on the underside **13** of the manhole cover (depending on the position of the ribs **17** or other structures on the manhole cover).

In the preferred configurations, the through-hole **15** (and consequently the connection **21**) is a generally central connection. The generally central connection **21** is located at the centre or towards the centre of the device and/or manhole cover body **10**. In other configurations the connection may be off-centre.

As the hollow connection shaft **23** is inserted/passes through the through-hole **15** of the manhole cover body **10**, the antenna unit **20** rests against the top surface **11** of the cover. The antenna unit base plate **22** is a flat structure providing a large bearing surface for engaging the top surface of the cover and provide a stable connection between the device and the cover once the antenna unit is secured to the cover (e.g. with a nut tightened on the underside of the cover).

The hollow connection shaft **23** is an elongate shaft structure generally extending along the longitudinal axis (Y) of the remote sensing device as shown in FIG. **5**. The elongate shaft structure extends along a longitudinal connection axis (C) passing through the through-hole **15**, as shown in FIG. **11**.

Preferably, the hollow connection shaft **23** is a rigid structure formed from durable material such as metal (stainless steel/ferrous material) or other suitable material known to a person skilled in the art.

Most preferably, the hollow connection shaft **23** is a threaded structure (comprises an external thread). The hollow connection shaft **23** is at least partially threaded to receive and secure the units **20**, **50** with a securing element (e.g. large securing nut).

In the preferred configurations, the connection **21** is the primary/main connection for securing the device onto the manhole cover **10**. The connection **21** formed between the antenna unit **20** and the monitoring unit **50** rigidly connects the device **100** with the manhole cover body **10** without any or with limited modification (formation of a through-hole) to traditional manhole covers. This can allow simple, quick and cost-efficient installation of the device **100**, even on-site at the manhole on the side of the road.

Preferably, the hollow connection shaft **23** depends from a lower surface of the base plate **22** of the antenna unit **20**, as shown in FIG. **6**.

In some configurations, the hollow connection shaft **23** is connected to the antenna unit **20**. The antenna unit **20** and the connection shaft **23** can be rigidly coupled/locked together e.g. using fasteners (countersunk screws), mating features (e.g. threaded components) or other locking engagement features known to a person skilled in the art.

In other configurations, the hollow connection shaft **23** is integrally formed with the antenna unit **20** (i.e. formed as one piece).

In one configuration, the connection **21** comprises one or more securing elements/nuts **24**, **25** for securing the antenna unit **20** and/or monitoring unit **50** to a structure of an infrastructure being monitored (e.g. to the manhole cover as shown in FIG. **10**, catch pit grate as shown in FIG. **16**, pipe of a raingarden as shown in FIG. **17**, or a structure on a trough, tank etc.).

After the hollow connection shaft **23** has been inserted through the manhole cover, preferably, a primary antenna fastener **24**, inserted from the underside of the manhole cover, secures the antenna unit **20** to the cover body **10** (i.e. as the fastener is tightened, the antenna unit **20** is tightened against the top surface **11** of the cover.) The primary antenna fastener **24** provides a stable connection for securing the antenna unit **20** to the manhole cover after it has been aligned/been calibrated to its desired operating position.

Preferably, after the antenna unit **20** has been secured to the manhole cover, the monitoring unit **50** is secured to the hollow connection shaft. In one configuration, a primary unit fastener **25** is inserted from the underside of the manhole cover to provide a rigid connection as it secures the monitoring unit **50** to the cover body **10** (i.e. as the fastener is tightened, the monitoring unit **50** is tightened upwards towards/against the bottom surface **12** of the cover.) As the primary unit fastener **25** is tightened the unit is clamped/sandwiched between the manhole cover (above the unit) and primary unit fastener **25** (below the unit).

In one configuration, the connection assembly **21** comprises a double fastener stack (i.e. including both the primary antenna fastener **24** and the primary unit fastener **25**) to connect both the antenna unit **20** and the monitoring unit **50** to the manhole cover body **10**.

In the preferred configurations, the antenna unit **20** is secured to the manhole cover body **10** before the monitoring unit **50** is secured to the manhole cover body with a fastener. Securing the antenna unit **20** before securing the monitoring unit **50** can improve the ease of installation.

Further, using separate fasteners **23**, **24** for securing each of the units **20**, **50** can improve the ease and accuracy of positioning the units **20**, **50**. One or both of the antenna unit **20** and monitoring unit **50** are free to rotate as desired to their operating position (e.g. the monitoring unit **50** can be position for the sensor to best detect the parameter) before securing the unit using the fastener.

In the preferred configurations, the primary connection fasteners **23**, **24** used to secure the units **20**, **50** to the hollow connection shaft **23** are nuts. The dimensions of the nut are preferably sufficiently large to secure the antenna unit **20**/monitoring unit **50** tight against/towards the cover body **10** (e.g. nuts having a M16 and/or M20 thread may be used). The securing nut preferably comprises a sufficient large engaging surface to secure against the monitoring unit **50** and/or cover body (component of infrastructure) to secure the units to form a rigid/robust connection.

In other configurations, other suitable fasteners known to a person skilled in the art may be used. Alternatively, other securing elements/techniques such as threading the units directly to the manhole cover or welding the unit(s) to the manhole cover may be used. In some configurations, an external thread of the hollow connection shaft engages directly with the component of infrastructure. For example, the threaded shaft is installed into a threaded recess in a manhole cover.

In some configurations, the connection assembly **21** further comprises components to improve the connection performance.

In the preferred configurations, the connection **21** further comprises washers or gaskets (e.g. anti-vibration washers) or plates **26** to evenly distribute the load of the fastener connection and/or reduce loosening of the connection due to vibrations. In some configurations, one of the plates **26** is an upper plate extending radially from a central hub **55** (further than typical washers) to improve reduce transfer of vibrations, space the monitoring unit **50** from the bottom surface **12** of the manhole cover **10** and/or accommodate other mounting elements.

In some configurations, the connection **21** further comprises an anti-crush element between the primary connection fasteners **23**, **24**, to limit how far the primary unit fastener **24** can press against the monitoring unit **50** (to reduce the likelihood of damaging the housing due to over tightening).

In some configurations, the remote sensing device comprises a secondary connection to connect the monitoring unit **50** to the central connection **21** (not shown). The secondary connection can supplement the primary connection **21**.

The secondary connection may be a threaded, snap-on, twist-fit, bayonet mount (FIG. **14**) or similar connection between the monitoring unit **50** and the hollow connection shaft **23** and/or the manhole cover body **10**.

Optionally, this secondary connection is less rigid in comparison to the primary connection discussed above. In some configurations, the secondary connection is compliant.

The remote sensing device **100** comprises a cable connection **40** for connecting the antenna unit **20** and the monitoring unit **50** to provide electrical, data and/or power communication between the units. In some configurations, the cable connection **40** is suitable for RF signal use (e.g. RG316 coaxial cable/MMCX connector may be used).

In some configurations, flexible or rigid cables may be used to connect the units together.

Preferably, the cable connection **40** to the antenna unit **20** is potted to avoid disconnection due to vibration.

Preferably, the cable connection **40** passes through the hollow connection shaft **23**, as shown in FIGS. **3** and **6**. The cable connection **40** is protected by the hollow connection shaft sleeve **23** and the cable is guided from the antenna unit **20** towards a connection point on the monitoring unit **50**.

In some configurations, the hollow connection shaft **23** comprises a side cut-out **28**. The side cut-out is configured to allow the cable connection **40** to exit the connection **23** laterally and connect with the monitoring unit **50**. The cable connection exits substantially laterally so that the cable is able to connect to the monitoring unit **50**, located adjacent/to a side/laterally of the hollow connection shaft.

It should be appreciated, the side cut-out **28** can help provide a simple to connect, and low-profile connection from the hollow connection shaft **23** and the monitoring unit **50** located radially outwards from the central hub **55**.

The side cut-out **28** can also be used as a positioning feature as the appropriate monitoring unit **50** position relative to the antenna unit **20** can be guided by the side-cut out for the cable connection **40**.

In other configurations, the antenna unit **20** and monitoring unit **50** comprises component-to-component pin connectors, i.e. not requiring a separate/external cable to connect components.

In some preferred configurations, components of the antenna unit **20** and the monitoring unit **50** connection are coaxially aligned to simplify connection between the antenna unit and the monitoring unit. Preferably, the central connectors are coaxially aligned with the through-hole **15** of the manhole cover. Simplifying connections can reduce the time of installation of the components as so they can be efficiently retrofitted to an existing manhole cover or installed onto the manhole cover during manufacture.

As the antenna unit **20** and monitoring unit **50** are separate and individually watertight units, a physical and electrical connection need to be made between the units. A physical connection supports the monitoring unit **50** and connect it to the antenna unit **20** (to form a rigid connection and resist impact and vibrations due to pedestrians and vehicles passing above). Preferably, as mechanical coupling is formed, the connection is physically so that the units are electrically linked, power and/or information can flow between the components of the smart manhole cover.

In some configurations, all necessary electrical connections are made by simply fitting/connecting the antenna unit **20** and the monitoring unit **50** together. As the mechanical connection(s) between the units **20**, **50** is established, an electrical connection is formed between the antenna unit and monitoring unit at the same time. The electrical connection and the mechanical connection are formed in one motion to connect the antenna unit and monitoring unit (e.g. both a physical and electrical connection is formed when monitoring unit is pressed and/or twisted onto the antenna unit) as shown in FIG. **15**.

Preferably, the electrical connection **21'** between the units is supported by (e.g. buried inside) the structural connection **21** between the units.

In some configurations, the antenna unit **20** and the monitoring unit **50** are connected by one or a combination of push-and-twist, press, snap, twist-fit, magnetic, bayonet mount (as shown in FIG. **14**) or similar connections. A manual connection of a separate cable connection **40** is not required, as aligning and physically connecting the antenna unit **20** and the monitoring unit **50** simultaneously forms the electrically, power and/or information connection.

Components Surrounding Hollow Connection Shaft

As previously described, a typical manhole cover has limited space on the underside of the cover (due to a limited cavity height **16** and obstructions e.g. ribs **17**). Efficient use of the limited space to provide a low-profile and robust device, while also having the desired monitoring/communication performance requirements is desirable.

A low-profile device integrated with the manhole cover can provide a simple smart manhole cover **1**, as all the components are connected to the cover (i.e. monitoring unit/sensor does not need to be installed down inside the manhole e.g. on the manhole wall). Further, the monitoring unit **50** on the underside will not be damaged when the manhole cover is dragged or placed on the ground.

In the preferred configurations, the monitoring unit **50** comprises an opening **28** for receiving the hollow connection shaft **23** as best shown in FIGS. **3** and **7**. Most preferably, the opening **28** allows the hollow connection shaft **23** to pass completely through the housing **51** of the monitoring unit **50**. The electrical components being housed within the water-tight/sealed monitoring unit **50** surrounding the opening **28** and hollow connection shaft **23** when assembled to protect internal components.

It should be appreciated the monitoring unit **50** with an opening **28** can work in synergy with the elongate connection shaft **23** feature to provide a low-profile remote sensing device **100** while also providing a strong/robust connection between the units and to attach the monitoring unit **50** to the manhole cover itself.

Preferably, the electrical components are located in a cavity **54** of the monitoring unit, arranged to be spaced radially around the opening as shown in FIG. **8**.

In the preferred configurations, the monitoring unit **50** comprises a hub **55**, the hub for mounting the elongate shaft. The hub **55** being a region of the monitoring unit surrounding the opening. Preferably, the hub **55** is isolated from the cavity (i.e. it is a region which does not house the electrical component). Preferably, the hub **55** is formed at or towards an inner perimeter of the monitoring unit **50** (around the opening **28**).

In these configurations, preferably all the electrical components on the underside **13** of the manhole cover **10** are housed in the monitoring unit **50** such that they are arranged to be spaced radially around the longitudinal connection axis (C) passing through the through-hole **15** of the manhole cover body **10** as shown in FIGS. **3** and **11**.

The opening **28** is aligned with the longitudinal connection axis (C) passing through the through-hole so that the connection **21** can pass through the opening. Preferably, the opening **28** is located generally centrally relative to the manhole cover body **10**. In other configurations, the opening **28** may be off-center.

The monitoring unit **50** may comprise one of the following housing profiles: a donut, ring, partial-ring, horseshoe or similar profile with an opening, when viewed in plan view. It should be appreciated the monitoring unit **50** can have other profiles (e.g. circular, semi-circular, square, rectangular, triangular, bean, peanut shaped etc.) with an opening **28**.

In plan view, the monitoring unit **50** and the electrical components it houses are not present in the region of the opening **28**. No active components are located in the opening region **28**, so that the hollow connection shaft/hub **28** can pass through the monitoring unit.

A low-profile can be achieved as the electrical components are spaced radially outwards from the hollow connection shaft **23** to use the space surrounding the connection to minimise the height of the device.

In these configurations, the monitoring unit **50** (and the electrical components it houses) has a low-profile and are located in a region between the top and bottom ends of the hollow connection shaft **23**. The monitoring unit **50** does not increase the height of the remote sensing device. The monitoring unit **50** is not located below the bottom end of the hollow connection shaft **23**.

A robust connection can be formed as the hollow connection shaft **23** passes through the opening **28** of the monitoring unit and a hub fastener (e.g. a securing nut) can be used to tightened the fastener against the monitoring unit **50**.

Further, the opening **28** of the monitoring unit **50** preferably provides a drainage pathway for water to prevent pooling on the housing (and to pass through the monitoring unit). Drainage pathways can direct water away from connection points or other potential regions of entry into the housing **51**. It should be appreciated the remote sensing device **100** operates in an environment where a lot of water can be present (e.g. due to rain and/or vapour or condensation build up). Consequently, potential leak pathways into the unit should be eliminated or minimised.

Preferably, the monitoring unit comprises one or more drain ports **56**. The drain ports being located on the hub **55** for water to pass through the monitoring unit **50**, without entering the cavity **54** housing the electrical components as best shown in FIG. **8**. In some configurations, the drain ports **56** are located on an upper housing portion **58** of the monitoring unit **50**. The drain ports **56** are spaced radially around the hollow connection shaft **23**/opening **28**. The drain ports **56** can form part of the drainage pathway for water to exit from a top side of the monitoring unit to a bottom side without pooling.

The opening **28** provides a drainage pathway, as water which enters from the manhole cover can flow around the antenna unit **20**, and downwards through the monitoring unit **50** (and out through the drain ports **56**) due to gravity. This reduces the likelihood of water leakage into the monitoring unit **50** as a connection point is not located on the top surface of the unit, and pooling on top of the monitoring unit is minimised.

Preferably, the opening **28** provides the primary drainage pathway for water to pass/exit through the monitoring unit **50**.

In some configurations, the opening **28** comprises a width/diameter greater than the hollow connection shaft **23** diameter. The opening **28** provides a pathway for water to flow around the hollow connection shaft **23**.

Monitoring Device Housing and Water Tightness

In some configurations, the monitoring unit **50** housing **51** comprises an upper housing portion **58** and a lower housing portion **59**, as best shown in FIG. **8**. The housing portions **58**, **59** are secured together to form an independent watertight housing connectable to the antenna unit **20**.

Preferably, the upper housing portion **58** has downwards facing shell profile to protect the electrical components of the monitoring device **50**. In these configurations, the lower housing portion **59** can be a substantially flat housing portion/a lid for closing the housing. The upper housing portion **58** and the lower housing portion **59** can be connected together with a plurality of fasteners (e.g. socket screws).

Preferably, the sealing surface between the upper housing portion **58** and the lower housing portion **59** is located on an underside of the monitoring unit to reduce the likelihood of ingress of water.

Gaskets **57** (shown in FIG. **8**) can be used to mechanically seal the space between the mating surface between the housing portions **58**, **59** and to minimise the likelihood of leakage into the monitoring unit **50**. Suitable gaskets known to a person skilled in the art may be used such as compression gaskets, neoprene gaskets, foam or liquid gaskets, RTV silicone/rubber gaskets etc.

In some configurations, a single gasket is provided between the upper housing portion **58** and the lower housing portion **59** to seal the mating surface between the housing portions. This is possible with certain monitoring unit profiles, such as in configurations where the monitoring unit has a horseshoe/partial ring profile.

Preferably, the monitoring units **50** are formed from polymers and additives/engineered plastics to suitable for enduring the environment of a manhole. It is anticipated that the monitoring units **50** can be formed from other materials or a combination of materials known by a person skilled in the art to be suitable for the environment the remote sensing device **100** will be in.

Details of Electrical Components

A simplified diagram of a preferred configuration showing possible components of the remote sensing device **100**, some of which are housed within the monitoring unit **50**, is shown in FIG. **13**.

In the preferred configurations, the remote sensing device **100** comprises a sensor **30** for detecting a parameter. The sensor **30** is one of the electrical components housed within the monitoring unit **50**.

In the preferred configurations, the remote sensing device **100** comprises a wireless communication module **31** located within the monitoring unit **50**. The wireless communication module **31** comprises a wireless transmitter and/or wireless receiver connected to the antenna.

The antenna preferably transmits information obtained by the sensor **30** to a remote receiver. For example, data from the device can be transmitted to a mobile phone, tablet or computer. Preferably, the device uses low-power wide-area network technologies (LPWAN). Optionally dedicated gateways can be used if necessary.

In the preferred configurations, the remote sensing device **100** comprises a controller **32** connected to the sensor **30** and wireless communication module **31**.

In the preferred configurations, the electrical components are connected on a PCB board **36**.

Optionally, the remote sensing device **100** comprises supplementary components such as a solar panel **34** or display **35** for displaying information from the device.

The sensor **30** is configured to detect a parameter associated with the manhole, manhole cover or other water infrastructure and/or their environment. The sensor **30** eliminates or at least reduces the frequency required for operators to check and/or access the manhole.

In some configurations, the sensor **30** is one of the follow types of sensors:

a) water level sensor;
b) non-contact flow sensor;
c) gas sensor;
d) temperature sensor;
e) moisture sensor;
f) tamper sensor;
g) vibration sensor; or
h) light sensor.

It is anticipated that in other configurations, other sensors types known to a person skilled in the art may be provided in the remote sensing device **100** for detecting other parameters as required.

In the preferred configurations, the sensor(s) is located within the monitoring unit **50** and the sensor does not protrude out of the housing **51**.

In contrast, a sensor or probe which extends or hangs below the manhole cover body **10**, into the manhole cavity may be unfavourable as the cover is not as compact/low-profile, and the sensor/probe may be damaged when the manhole cover **1** is being installed/moved/dragged. Furthermore, sensor(s) integrated with the manhole prevents a person from needing to enter the manhole to install/maintain/replace the sensor.

In the preferred configurations, the sensor **30** is a radar sensor for monitoring water level. The radar sensor emits electromagnetic waves to monitor water level in the manhole. It should be appreciated, a radar sensor can provide advantages in a manhole/water infrastructure application as it can provide consistent, accurate readings (as it is not affected by disruptions such as wind or airflow) and uses low power. The radar sensor can also interpret different interface layers in the measurement environment. Therefore, the radar sensor can see through/identify different media including dust, wood, leaves, oil, foam on water.

The radar sensor **30** comprises a lens **35** (e.g. a Fresnel lens) as shown in FIG. **9**. In the preferred configurations, the lens **35** is integrated with the housing of the monitoring unit **50** (preferably, integrated with the lower housing portion/lid **59**). The lens focal length may be adjusted as required by changing the lid **59**.

It should be appreciated integration of the lens **35** with the monitoring unit housing **51** can provide advantages such as reducing potential leak pathways and/or water ingress failure points, providing a low-profile/compact device, reduces production costs and complexity compared to a lens manufactured as part of the PCB or otherwise, and also allowing for simple lens replacement for different applications.

In the preferred configurations, the monitoring unit **50** comprises a water rejecting region **36**. The water rejecting region **36** is preferably on the housing **51** of the monitoring unit, in a focal area of the lens **35** as shown in FIG. **9**.

In some configurations, the water rejecting region **36** is a ramp on the housing surface for draining water away from the focal area/lens view of the radar sensor **30** as best shown in FIGS. **2** and **8**. The ramp **36** drains downwards and away from the focal area/view of the lens due to gravity. In the pictured configuration, the ramp **36** drains downwardly and inwardly towards the opening **28**. In other configurations, the ramp **36** drains downwardly and outwardly away from the opening **28**

Stagnate water/moisture due to rain or vapour on the housing in the focal area of the radar sensor is undesirable as it can cause interference with the signal and prevent radar from seeing various interface zones.

Optionally, the ramp/water rejecting region **36** comprises a hydrophobic coating or other features or coatings to promote water to move away from the focal area of the sensor.

Dimensions

It is anticipated that components of the remote sensing device **100** are sized and profiled as required for different manhole covers and/or different environments and situations. A range of suitable dimensions for some configurations for the components/structures of the remote sensing device **100** will now be provided, with referenced to FIGS. **4** and **6**.

In the preferred configurations the monitoring unit (**50**) comprises a width/diameter (**61**) between 50 mm and 200 mm.

Most preferably, the monitoring unit (**50**) comprises a width/diameter (**61**) between 50 mm and 160 mm.

In the preferred configurations the monitoring unit (**50**) comprises a height (**62**) between 25 mm and 60 mm.

Most preferably, the monitoring unit (**50**) comprises a height (**62**) between 30 mm and 40 mm.

Preferably, the monitoring unit (**50**) comprises a height less than the height of the cavity (**16**) on the underside (**13**) of the manhole cover body, so the manhole cover can rest on the ground/dragged without damaging the monitoring unit.

In the preferred configurations the antenna unit (**20**) comprises a width/diameter (**63**) between 50 mm and 160 mm.

Most preferably, the antenna unit (**20**) comprises a width/diameter (**63**) between 50 mm and 120 mm.

Preferably, the antenna unit **20** comprises a surface area big enough to receive the antenna wire and provide a good signal to communicate information to/from the smart manhole cover **1**.

In the preferred configurations the antenna unit (**20**) comprises a height (**64**) between 5 mm and 20 mm.

Most preferably, the antenna unit (**20**) comprises a height (**64**) between 5 mm and 15 mm.

Preferably, the antenna unit **20** comprises a flat profile so that it does not act as a hazard to vehicles or people above the manhole cover **1**.

In the preferred configurations the hollow connection shaft (**21**) has a height (**65**) between 15 mm and 60 mm.

Most preferably, the hollow connection shaft (**21**) has a height (**26**) between 15 mm and 40 mm.

Preferably, one or more of the connectors associated and/or within the monitoring unit **50** are horizontally arranged such that they are generally aligned with the horizontal plane of the manhole cover. Connectors in this orientation helps form a compact/low-profile cover

Modular Units

In some configurations, the electrical components are housed within two or more separate modular monitoring units **50** (FIG. **12**). Preferably, the one or more modular monitoring units **50** are removably coupled on the underside **13** of the cover body **10**.

Preferably, in these configurations, as multiple modular monitoring units **50**, **50'**, **50"** are physically connected, they become electrically linked, so that power and/or information can flow between the modular monitoring units. The electrical contacts **53** of neighbouring modular units are formed and broken, as the modular units are physically connected and disconnected from each other.

In the most preferred configurations, the modular units **50** are removably connected together. Removably connected modular units **50** may be advantageous as the modular units can be easily installed or removed for maintenance/repair or for providing and removing certain functionality as required.

Preferably, neighbouring modular units comprise complementary features for connecting the modular units together. In some configurations, the neighbouring modular units are connected together by one or a combination of the following connection types: threaded, plug-fit, magnetic, bayonet mount connections. It is anticipated other suitable connection features may be used to connect neighbouring modular units.

In these configurations, preferably the connectors are of a type such as spring-loaded or sliding connectors that aid in aligning the electrical contacts **53** between the modular units **50**, **50'**.

In some configurations, two or more of the modular units are connected laterally, so that the units are located within the cavity **16** on the underside of the manhole cover and to maintain a low-profile device.

In other configurations two or more of the modular units are connected in series in a vertical stack. In these configurations, preferably a bottom surface of an upper modular unit **50** is configured to engage with a top surface of a lower modular unit **50'**.

In the preferred configurations, at least two of the modular units house components of different types to provide different functions for the smart manhole cover.

For example, in one configuration, a first modular unit **50** houses a first sensor **30** to detect a first parameter, and a second modular unit **50'** houses a second sensor **30'** to detect a second parameter. In one configuration, the first modular unit **50** houses a tamper sensor to detect whether the manhole cover **1** has moved, and the second modular unit **50'** houses a water level sensor to detect the water level within the manhole.

In other configurations, two or more of the modular units comprise components of the same type to augment a function. For example, in one configuration, the smart manhole cover **1** comprises two power supplies. Each power supply **33** is housed in a separate modular unit **50**, **50'**.

It is anticipated that supplementary modular units may be connected to the smart manhole cover **1** to increase the different types of functions available or augment the functions of the smart manhole cover as desired.

To those skilled in the art to which the invention relates, many changes in construction and widely differing embodiments and applications of the invention will suggest themselves without departing from the scope of the invention as defined in the appended claims.

This invention may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, and any or all combinations of any two or more of said parts, elements or features, and where specific integers are mentioned herein which have known equivalents in the art to which this invention relates, such known equivalents are deemed to be incorporated herein as if individually set forth.

The invention claimed is:

1. A remote sensing device for monitoring a condition within a water infrastructure network comprising:

an antenna unit configured to be located at or towards a top surface of a water infrastructure component in the water infrastructure network, the antenna comprises a flat base for engaging with the top surface of the water infrastructure component, the antenna unit having an elongate connection shaft comprising a hollow depending therefrom, the elongate connection shaft providing a mounting structure on the underside of the water infrastructure component for receiving and supporting a structure;

a monitoring unit for housing electrical components including a sensor for detecting a parameter, the monitoring unit removably mounted to the antenna;

wherein the monitoring unit comprises a hub for mounting the monitoring unit to the elongate connection shaft, the hub being a region of the monitoring unit surrounding an opening of the monitoring unit, the opening configured such that the elongate connection shaft passes completely through the housing of the monitoring unit;

wherein the electrical components are located in a cavity of the monitoring unit, arranged to be spaced radially around the hub and the monitoring unit is located in a region between the top and bottom ends of the elongate connection shaft; and

wherein the antenna unit and monitoring unit are separate and individually water-tight structures.

2. The remote sensing device as claimed in claim **1** wherein the elongate connection shaft comprises an external thread for making an essentially rigid connection, and further comprises a nut associated with the threaded elongate connection shaft, the nut having a large engaging surface for engaging against the water infrastructure component, to form the essentially rigid connection.

3. The remote sensing device as claimed in claim **1** wherein the elongate connection shaft is connected directly to the water infrastructure component to form an essentially rigid connection, and

wherein an external thread of the elongate connection shaft engages directly with the water infrastructure component.

4. The remote sensing device as claimed in claim **1** wherein an electrical connection is formed between the antenna unit and monitoring unit at the same time as a mechanical connection between the units is established with one motion, and

wherein the electrical connection is a push type electrical connector incorporated with the elongate connection shaft.

5. The remote sensing device as claimed in claim **1** wherein the monitoring unit comprises one or more drain ports located on the hub for water to pass through the monitoring unit, without entering the cavity housing the electrical components.

6. The remote sensing device as claimed in claim **1** wherein in plan view, the monitoring unit comprises one of the following general housing profiles:

a) donut,

b) ring,

c) partial-ring, or

d) horseshoe.

7. The remote sensing device as claimed in claim **1** wherein the sensor is one of the following:

a) water level sensor,

b) non-contact flow sensor,

c) gas sensor,

d) temperature sensor,

e) moisture sensor,

f) tamper sensor,

g) vibration sensor, or

h) light sensor.

8. The remote sensing device as claimed in claim **1** wherein the sensor is a radar sensor comprising a lens, and wherein the lens is integrated with a housing of the monitoring unit, and

comprises a water rejecting region in a focal area of the lens, and

wherein the water rejecting region is a ramp on a housing surface of the monitoring unit for draining water downwards away from the focal area.

9. The remote sensing device as claimed in claim **1** wherein the electrical components housed in the monitoring unit comprise one or a combination of:

a) a controller,

b) a wireless communication module,

c) a power supply, and

wherein the electrical components are housed within one monitoring unit.

10. The remote sensing device as claimed in claim **1** wherein the monitoring unit comprises an upper housing portion and a lower housing portion, the upper housing portion comprises a downwards facing shell profile and the lower housing portion comprises a substantially flat lid profile, and

wherein a single gasket is provided between the upper housing portion and the lower housing portion to seal a mating surface between the housing portions.

11. The remote sensing device as claimed in claim **10**, wherein the water infrastructure component is a:

a) manhole cover,

b) grate,

c) cap, or

d) lid.

12. The remote sensing device as claimed in claim **1** wherein the monitoring unit is not located below the bottom end of the hollow connection shaft.

13. A smart manhole cover comprising:

a manhole cover body; and

a remote sensing device as claimed in claim **1**;

wherein the remote sensing device is connected to the manhole cover body such that the antenna unit is located at an upper surface of the manhole cover body and the monitoring unit is located on an underside of the manhole cover body.

14. The smart manhole cover as claimed in claim **13** wherein the elongate connection shaft passes through a through-hole of the manhole cover body for a connection between the antenna unit and monitoring unit, and

wherein the elongate connection shaft extends along a longitudinal connection axis passing through the through-hole, and

wherein the manhole cover body comprises a cavity on the underside of the cover for receiving the monitoring unit and the monitoring unit has a height less than the cavity height such that the monitoring unit does no protrude below a bottom surface of the manhole cover body.

15. A method of sensing a parameter comprising:

providing a remote sensing device as claimed in claim **1**;

monitoring a parameter;

wirelessly communicating data from the remote sensing device to a remote receiver.

16. The method of sensing a condition as claimed in claim **15** further comprising passing the elongate connection shaft through an opening of the monitoring unit and securing the monitoring unit with a hub fastener tightened against a surface of the monitoring unit.

17. The method of sensing a condition as claimed in claim **15** wherein the parameter is associated with a manhole and the remote sensing device is connected to a manhole cover body such that the antenna unit is located at an upper surface of the manhole cover body and the monitoring unit is located on an underside of the manhole cover body.

18. The method of sensing a condition as claimed in claim **15** further comprising securing the antenna unit to a manhole cover body before securing the monitoring unit.

19. The method of sensing a condition as claimed in claim **15** wherein the remote sensing device is retrofitted to an existing manhole cover body.

20. The method of sensing a condition as claimed in claim **15** wherein the remote sensing device is associated with and connected to one of the following:

a) sewers,

b) stormwater catch pits,

c) rain gardens,

d) water troughs,

e) gross pollutant traps, or

f) waste tanks.

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